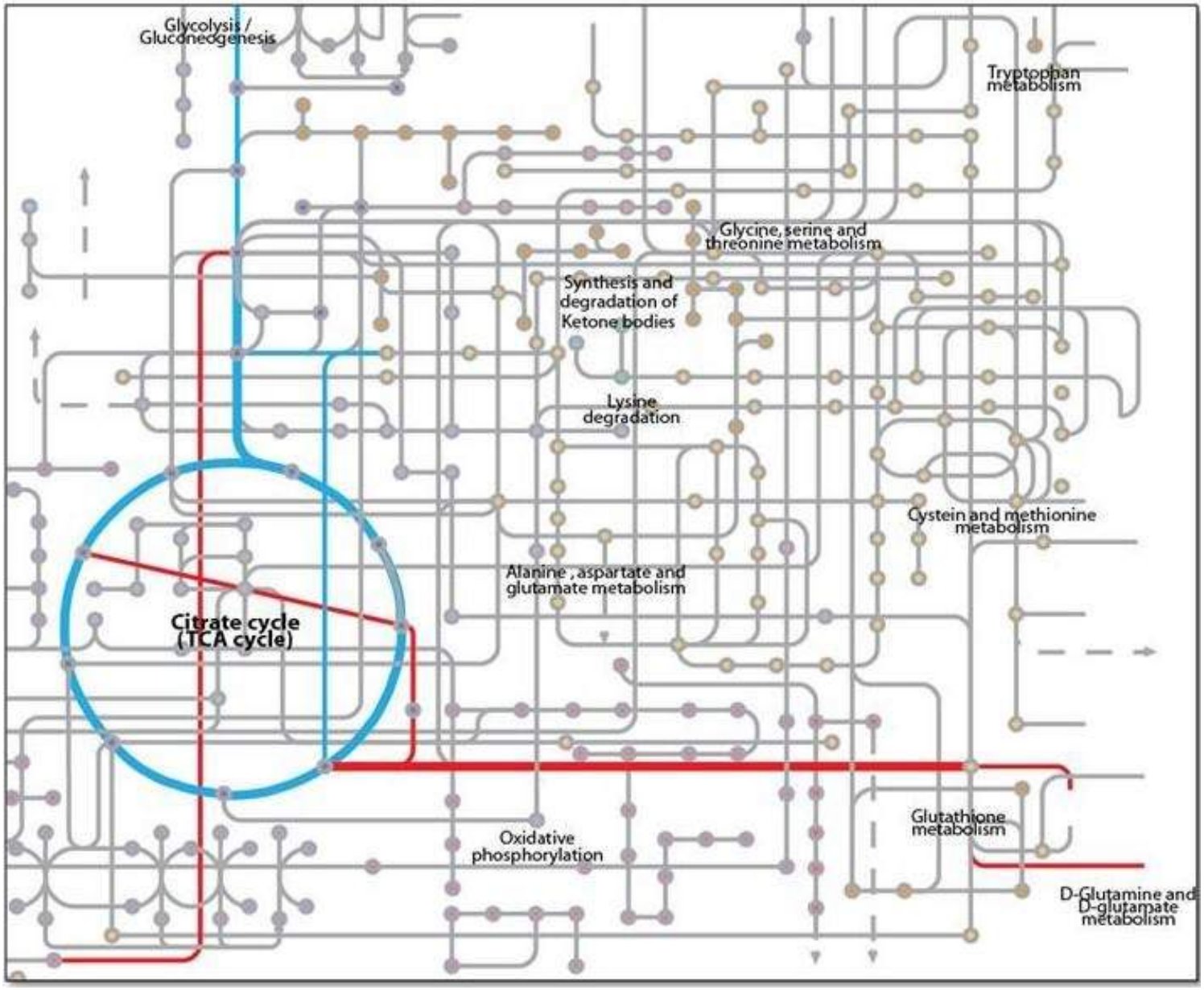
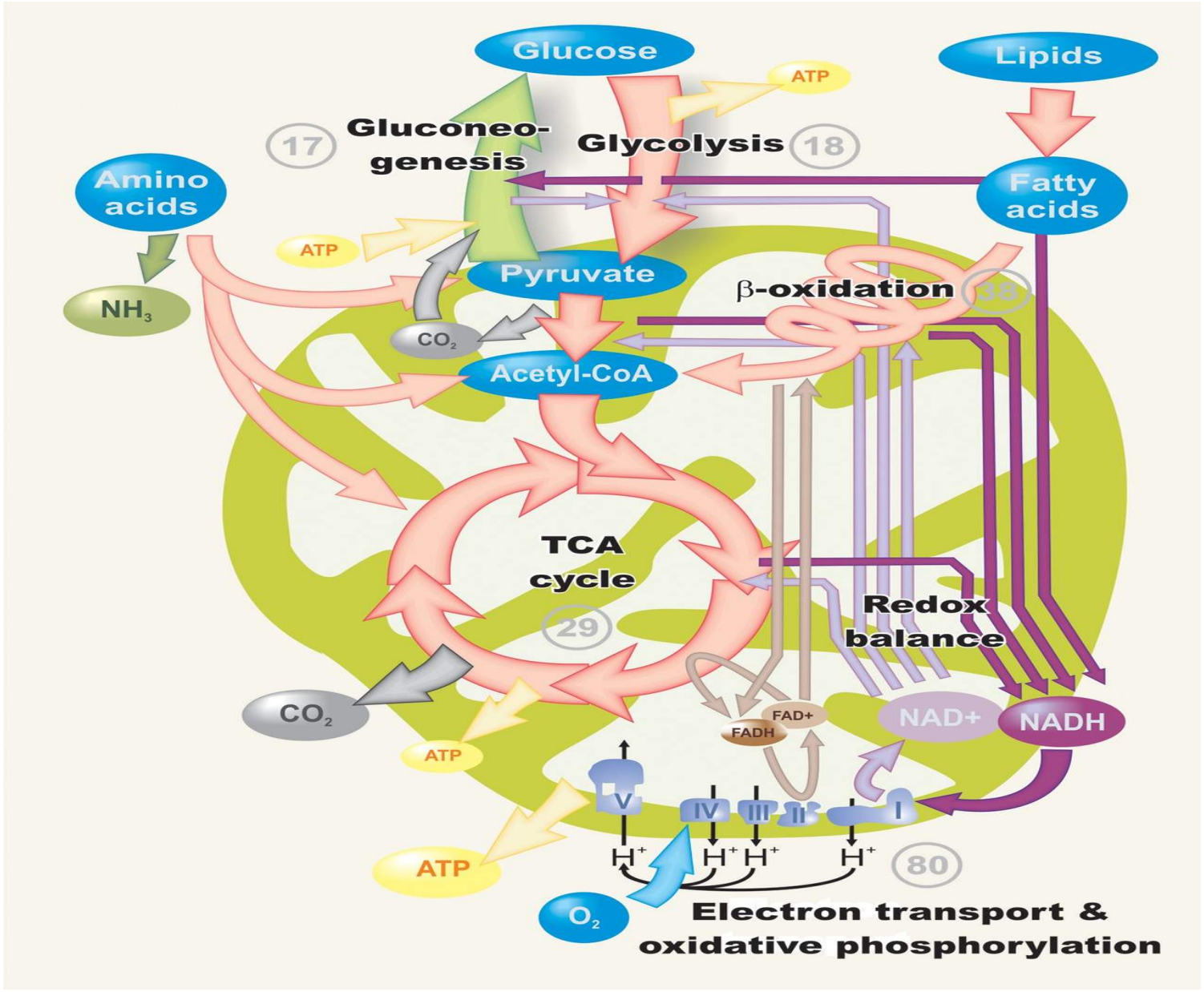
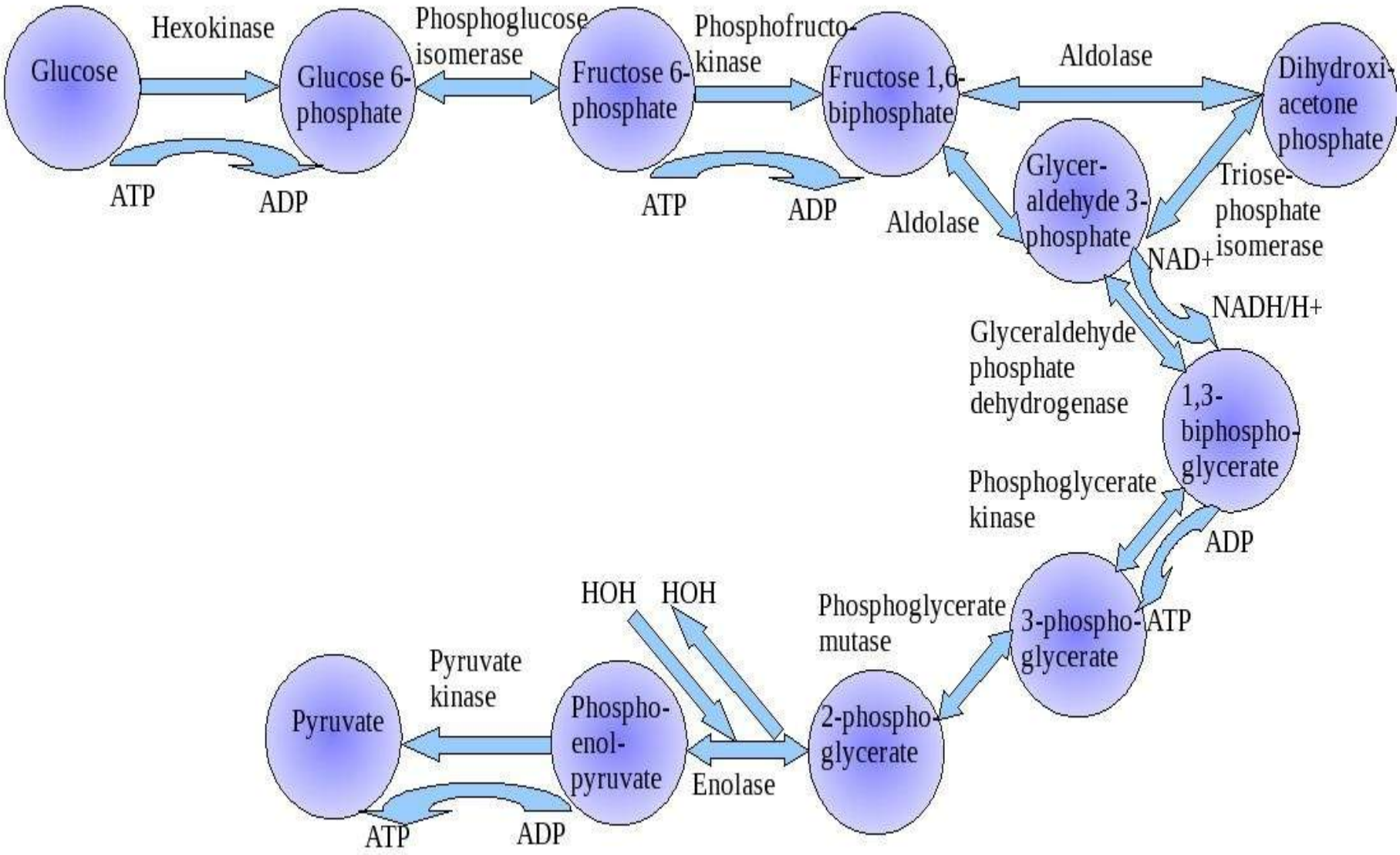
METABOLIC PATHWAYS




Glucose
Lipids
ATP
17
Gluconeo-
genesis
Glycolysis
18
Amino
acids
Fatty
acids
ATP
Pyruvate
β-oxidation
38
NH3
CO2
Acetyl-CoA
TCA
cycle
29
Redox
balance
CO2
FAD+
FADH
NAD+
NADH
ATP
V
IV
III
II
I
H+
H+
H+
H+
ATP
80
O2
Electron transport &
oxidative phosphorylation

GLYCOLYSIS


Glucose
Hexokinase
Glucose 6-phosphate
ATP
ADP
Phosphoglucose isomerase
Fructose 6-phosphate
Phosphofructo-kinase
Fructose 1,6-biphosphate
ATP
ADP
Aldolase
Dihydroxi-acetone phosphate
Aldolase
Glycer-aldehyde 3-phosphate
Triose-phosphate isomerase
NAD+
NADH/H+
Glyceraldehyde phosphate dehydrogenase
1,3-biphospho-glycerate
Phosphoglycerate kinase
ADP
ATP
3-phospho-glycerate
Phosphoglycerate mutase
HOH
HOH
2-phospho glycerate
Enolase
Phospho-enol-pyruvate
Pyruvate kinase
Pyruvate
ATP
ADP

Glycolysis: Derived from Greek words;

Glykys = Sweet, Lysis = splitting
During this process one molecule of glucose (6 carbon molecule) is degraded into two molecules of pyruvate (three carbon molecule).

Free energy released in this process is stored as 2 molecules of ATP, and 2 molecules of NADH.

Glucose + 2NAD+ = 2Pyruvate + 2NADH + 2H+ ΔG^o = -146 kJ/mol
2ADP + 2Pi = 2ATP + $2H_2O$ ΔG^o = 2X(30.5 kJ/mol) = 61 kJ/mol

----

ΔG^o (overall) = -146+61 = -85 kJ/mol

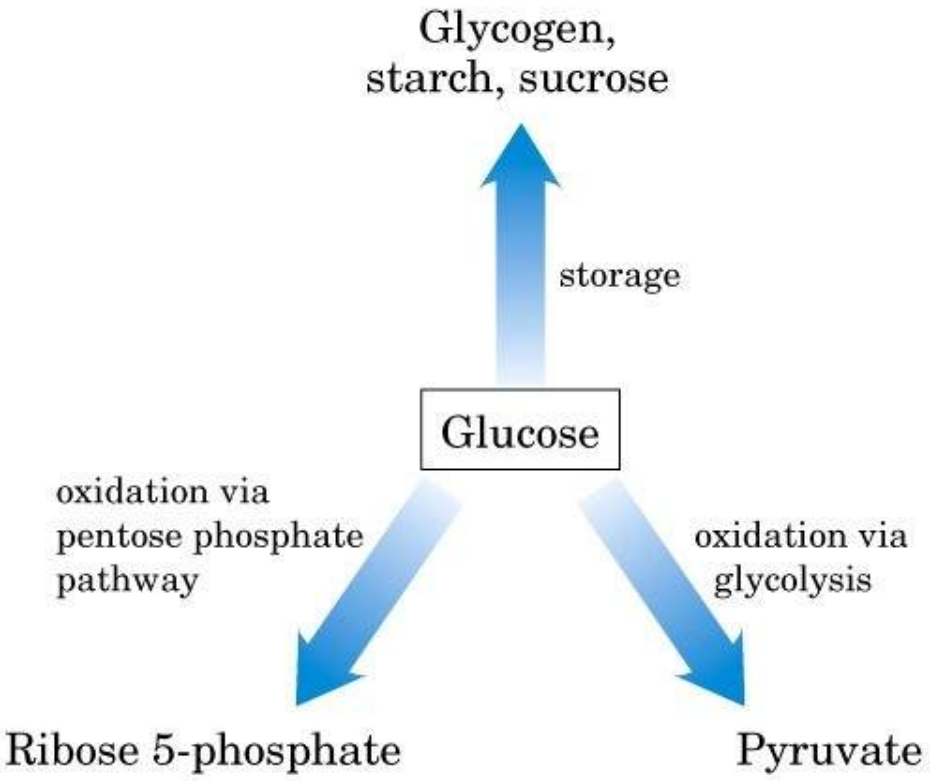
In standard condition glycolysis is an exergonic reaction which tends to be irreversible because of negative ΔG^o.

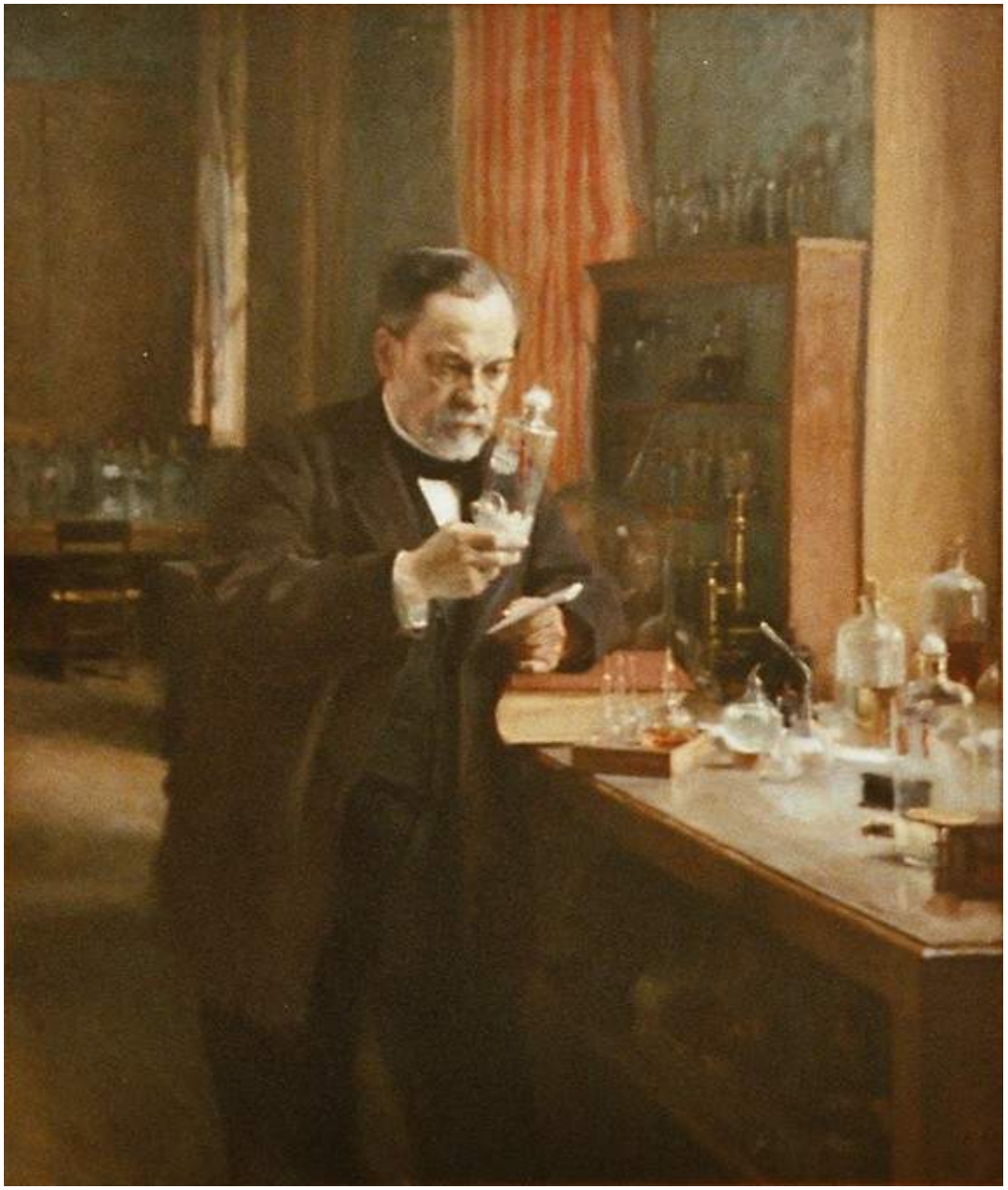
Fate of glucose in living systems

$$Glucose + 6O_2 = 6CO_2 + 6H_2O \quad \Delta G^o = -2840\ kJ/mol$$

$$Glucose + 2NAD^+ = 2Pyruvate + 2NADH + 2H^+ \quad \Delta Go = -146\ kJ/mol$$

5.2% of total free energy that can be released by glucose is released in glycolysis.





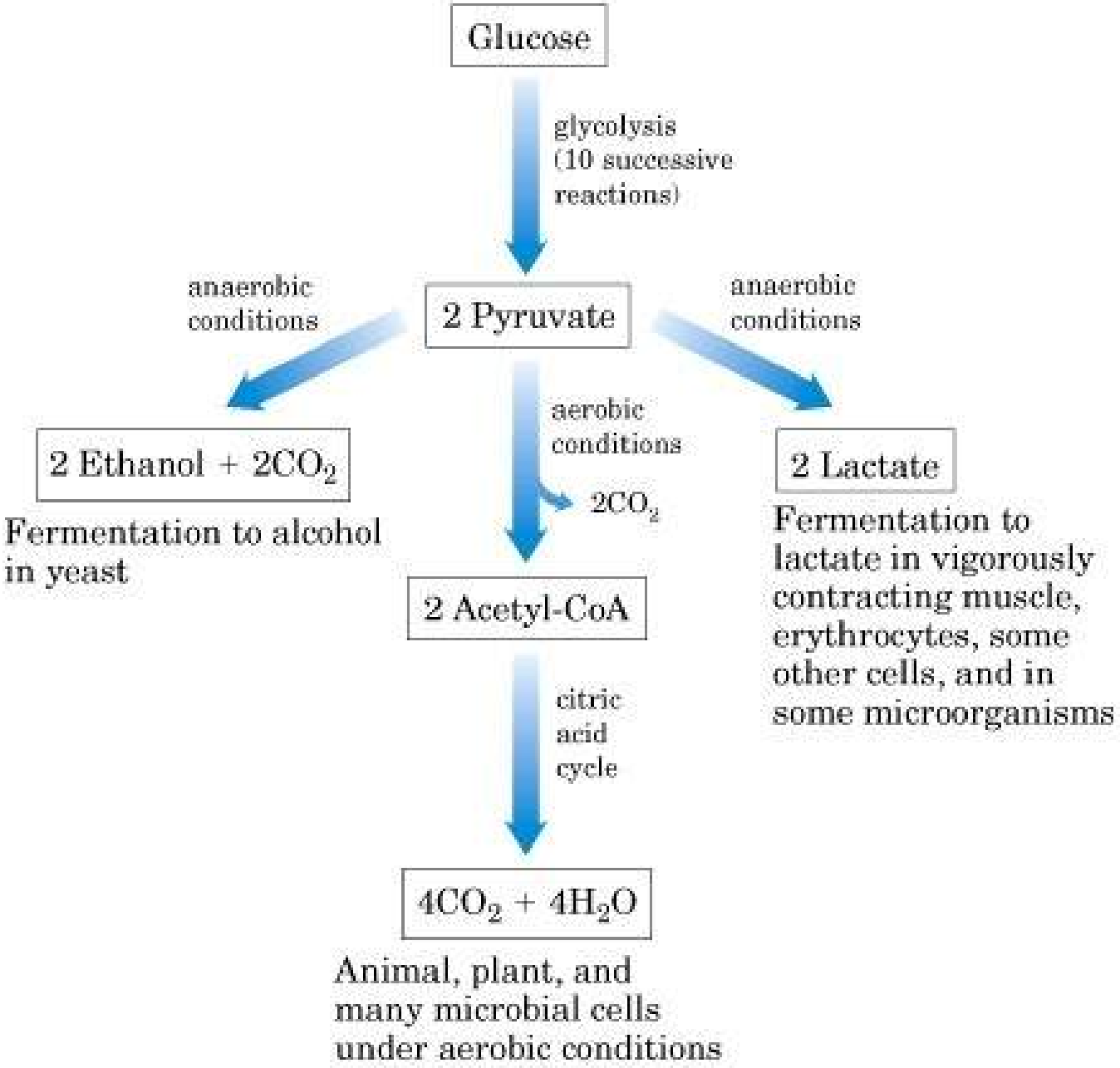
Louis Pasteur (1822-1895)

Historical Perspective

Glycolysis was the very first biochemistry or oldest biochemistry studied. It is the first metabolic pathway discovered.
Louis Pasture 1854-1864: Fermentation is caused by microorganism.
Pastuer's effect: Aerobic growth requires less glucose than anaerobic condition.
Buchner; 1897: Reactions of glycolysis can be carried out in cell-free yeast extract.

Harden and Young 1905: 1: inorganic phosphate is required for fermentation. 2: yeast extract could be separated in small molecular weight essential coenzymes or what they called Co-zymase and bigger molecules called enzymes or zymase.

Inhibitor studies: Iodoacetate treatment resulted in the accumulation of fructose 1,6biphosphate. Similarly fluoride caused accumulation of 2-phosphoglycerate and 3-phosphoglycerate.
1940: with the efforts of many workers, complete pathways for glycolysis was established.


Glucose
glycolysis (10 successive reactions)
2 Pyruvate
anaerobic conditions
anaerobic conditions
aerobic conditions
$2CO_2$
2 Ethanol + $2CO_2$
Fermentation to alcohol in yeast
2 Acetyl-CoA
2 Lactate
Fermentation to lactate in vigorously contracting muscle, erythrocytes, some other cells, and in some microorganisms
citric acid cycle
$4CO_2 + 4H_2O$
Animal, plant, and many microbial cells under aerobic conditions

There are 10 enzyme-catalyzed reactions in glycolysis. There are two stages

Stage 1: Reactions 1-5) A preparatory stage in which glucose is phosphorylated, converted to fructose which is again forphorylated and cleaved into two molecules of glyceraldehyde-3-phosphate. In this phase there is an investment of two molecules of ATP.

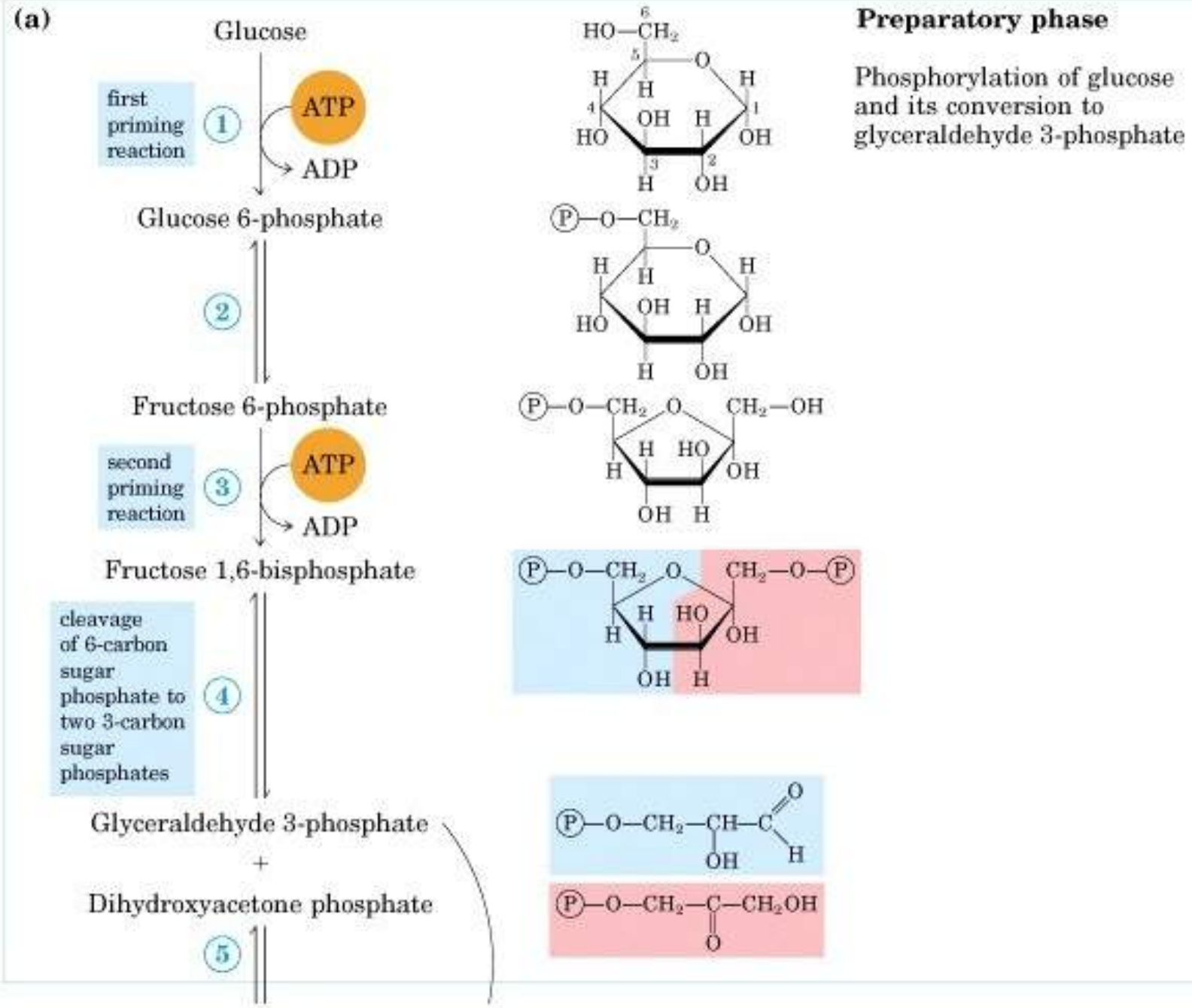
Stage 2: (reactions 6-10) The two molecules of glyceraldehyde-3-phosphate are converted to pyruvate with concomitant generation of four ATP molecules and two molecules of NADH. Thus there is a net gain of two ATP molecules per molecule of Glucose in glycolysis.

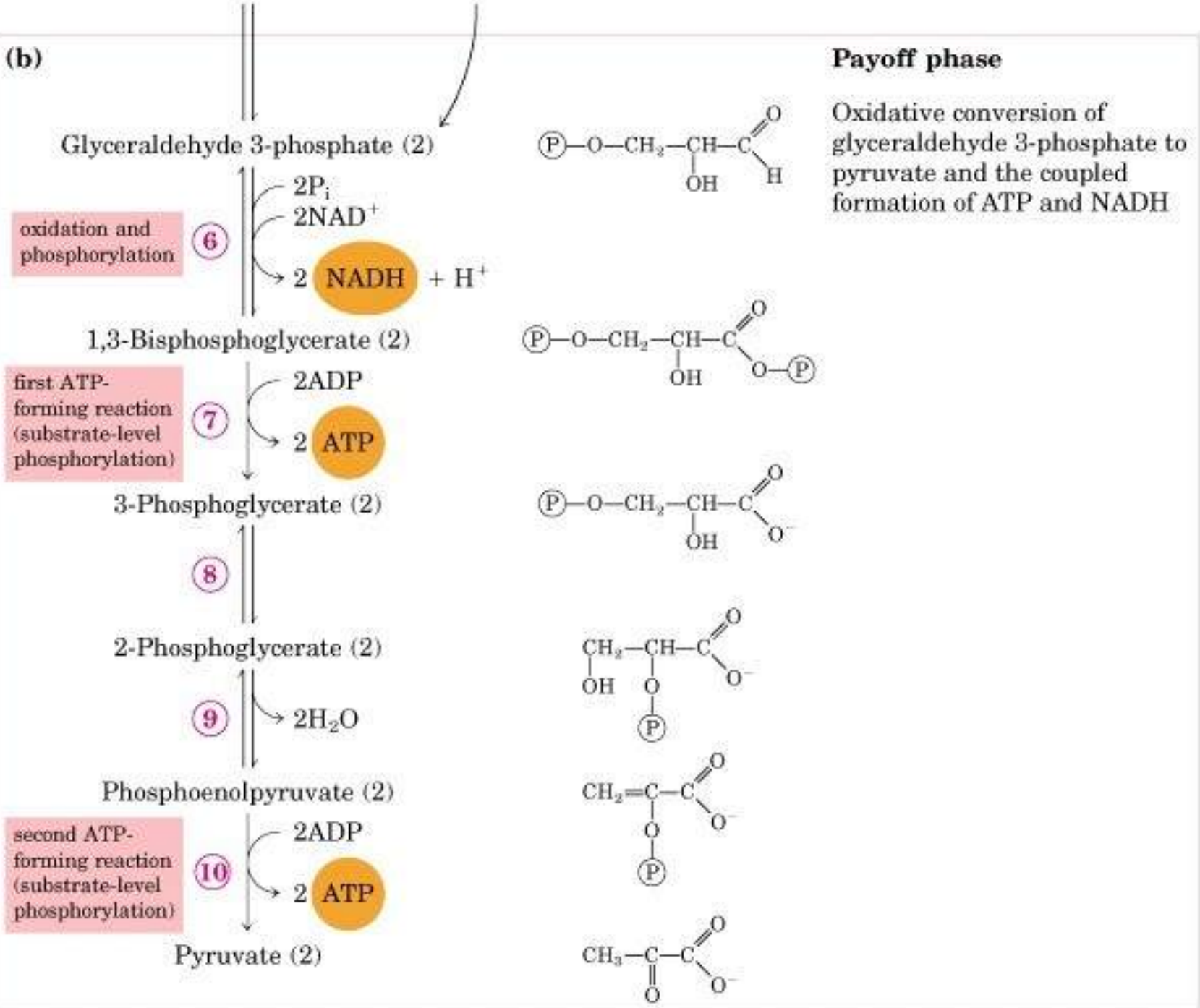
Importance of phosphorylated intermediates:

1. **Possession of negative charge which inhibit their diffusion through membrane.**

2. **Conservation of free energy in high energy phosphate bond.**

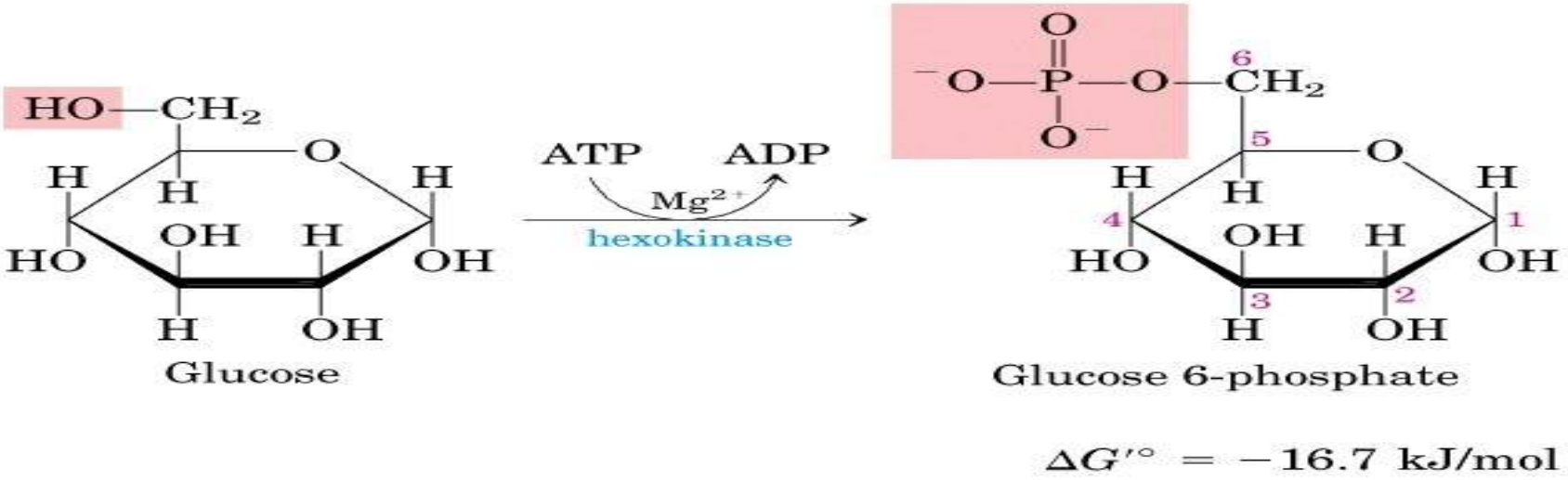
3. **Facilitation of catalysis.**


(a)
Glucose
first priming reaction
1
ATP
ADP
Glucose 6-phosphate
2
Fructose 6-phosphate
second priming reaction
3
ATP
ADP
Fructose 1,6-bisphosphate
cleavage of 6-carbon sugar phosphate to two 3-carbon sugar phosphates
4
Glyceraldehyde 3-phosphate
+
Dihydroxyacetone phosphate
5
Preparatory phase
Phosphorylation of glucose and its conversion to glyceraldehyde 3-phosphate


(b)
Payoff phase
Oxidative conversion of glyceraldehyde 3-phosphate to pyruvate and the coupled formation of ATP and NADH
Glyceraldehyde 3-phosphate (2)
2Pi
2NAD+
oxidation and phosphorylation
6
2 NADH + H+
1,3-Bisphosphoglycerate (2)
first ATP-forming reaction (substrate-level phosphorylation)
7
2ADP
2 ATP
3-Phosphoglycerate (2)
8
2-Phosphoglycerate (2)
9
2H2O
Phosphoenolpyruvate (2)
second ATP-forming reaction (substrate-level phosphorylation)
10
2ADP
2 ATP
Pyruvate (2)

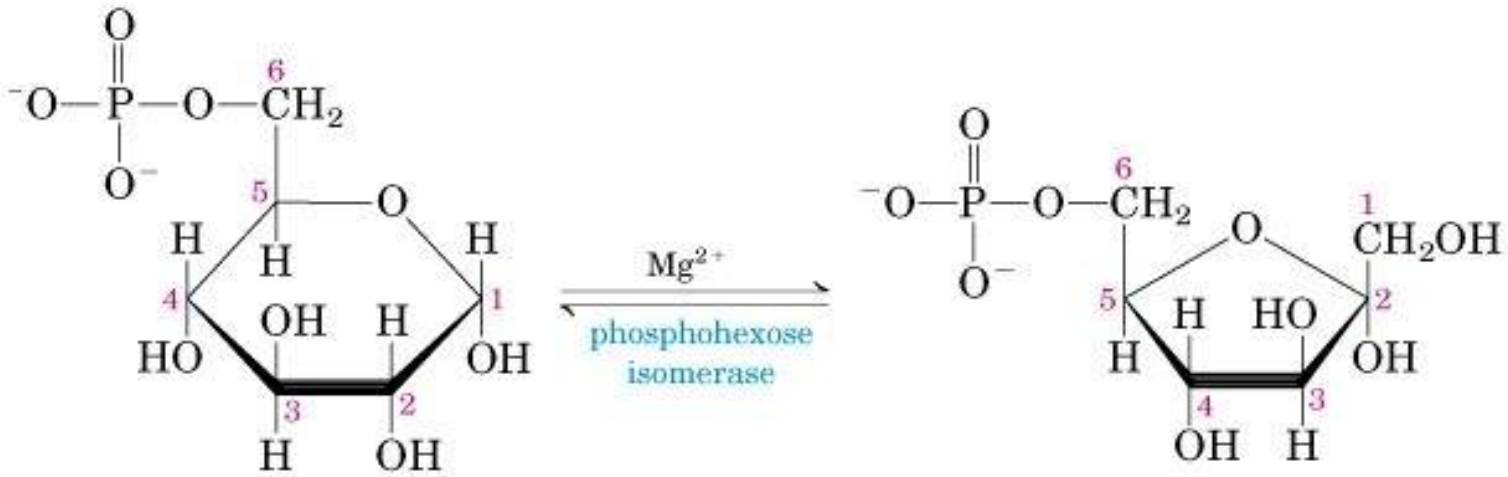
1. Hexokinase reaction: phosphorylation of hexoses (mainly glucose)

I. This enzyme is present in most cells. In liver Glucokinase is the main hexokinase which prefers glucose as substrate.
II. It requires Mg-ATP complex as substrate. Un-complexed ATP is a potent competitive inhibitor of this enzyme.
III. Enzyme catalyses the reaction by proximity effect; bringing the two substrate in close proximity.
IV. This enzyme undergoes large conformational change upon binding with Glucose. It is inhibited allosterically by G6P.



2. Phosphoglucose Isomerase or Phosphohexose Isomerase: Isomerization of G6P to Fructose 6 phosphate.

I. This enzyme catalyzes the reversible isomerization of G6P (an aldohexose) to F6P (a ketohexose).
II. This enzyme requires Mg^{++} for its activity.
III. It is specific for G6P and F6P.



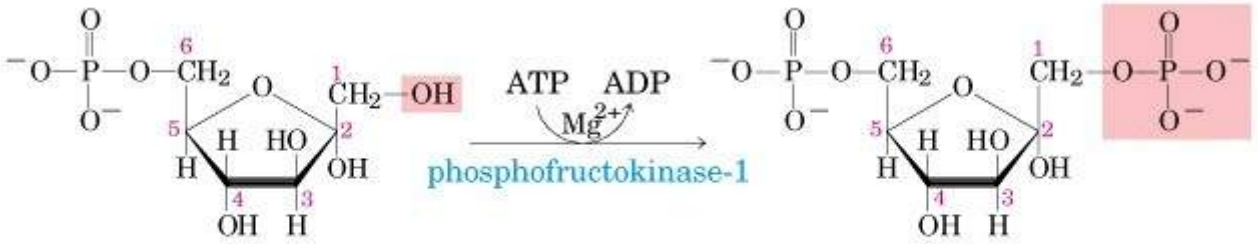
Glucose 6-phosphate

Fructose 6-phosphate

$\Delta G'^{\circ} = 1.7$ kJ/mol

3. Phosphofructokinase-1 Reaction: Transfer of phosphoryl group from ATP to C-1 of F6P to produce Fructose 1,6 biphosphate.

I. This step is an important irreversible, regulatory step.
II. The enzyme Phosphofructokinase-1 is one of the most complex regulatory enzymes, with various allosteric inhibitors and activators.
III. ATP is an allosteric inhibitor, and Fructose 2,6 biphosphate is an activator of this enzyme.
IV. ADP and AMP also activate PFK-1 whereas citrate is an inhibitor.



Fructose 6-phosphate

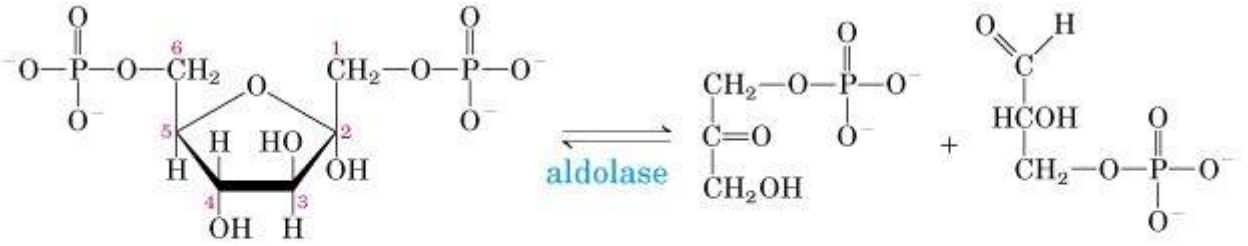
Fructose 1,6-bisphosphate

$\Delta G^{\circ} = -14.2$ kJ/mol

4. Aldolase Reaction: Cleavage of Fructose 1,6 biphosphate into glyceraldehyde 3 phosphate (an aldose) and dihydroxy acetone phosphate (a ketose).

I. This enzyme catalyses the cleavage of F1,6 biphosphate by aldol condensation mechanism.

II. As shown below, the standard free energy change is positive in the forward direction, meaning it requires energy. Since the product of this reaction are depleted very fast in the cells, this reaction is driven in forward direction by the later two reactions.



Fructose 1,6-bisphosphate

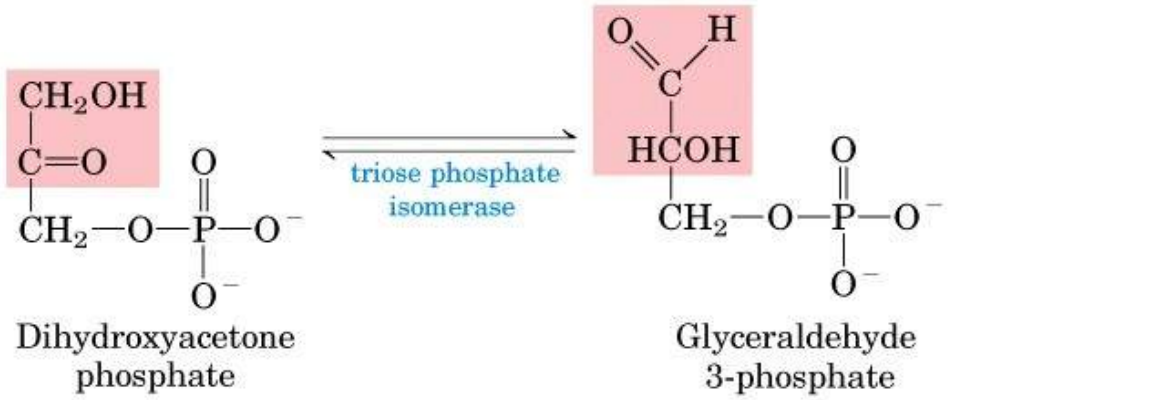
Dihydroxyacetone phosphate

Glyceraldehyde 3-phosphate

$\Delta G'^{\circ} = 23.8 kJ/mol$

5. Triose phosphate mutase reaction: Conversion of Dihydroxyacetone phosphate to glyceraldehyde 3 Phosphate.

I. This a reversible reaction catalysed by acid-base catalysis in which Histidine-95 and Glutamate -165 of the enzyme are involved.



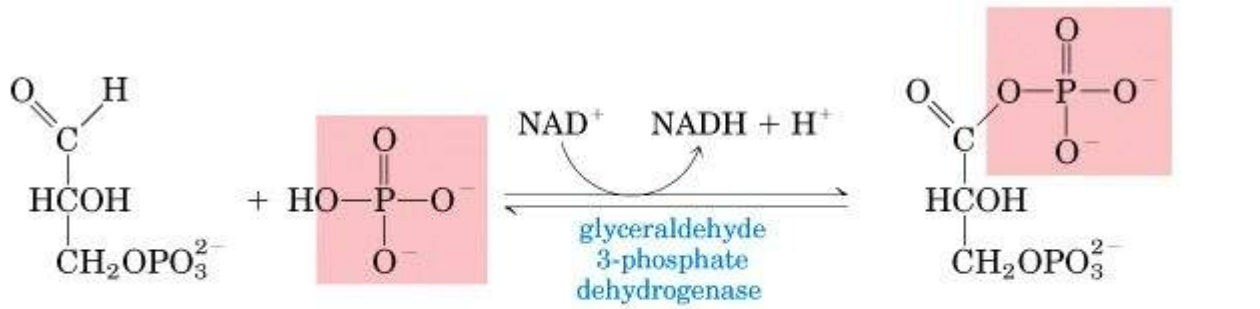
$\Delta G'^{\circ} = 7.5$ kJ/mol

6. Glyceraldehyde-3-phosphate dehydrogenase reaction (GAPDH): Conversion of GAP to Bisphosphoglycerate.

I. This is the first reaction of energy yielding step. Oxidation of aldehyde derives the formation of a high energy acyl phosphate derivative.

II. An inorganic phosphate is incorporated in this reaction without any expense of ATP.

III. NAD^+ is the cofactor in this reaction which acts as an oxidizing agent. The free energy released in the oxidation reaction is used in the formation of acylphosphate.



$\Delta G'^\circ = 6.3$ kJ/mol

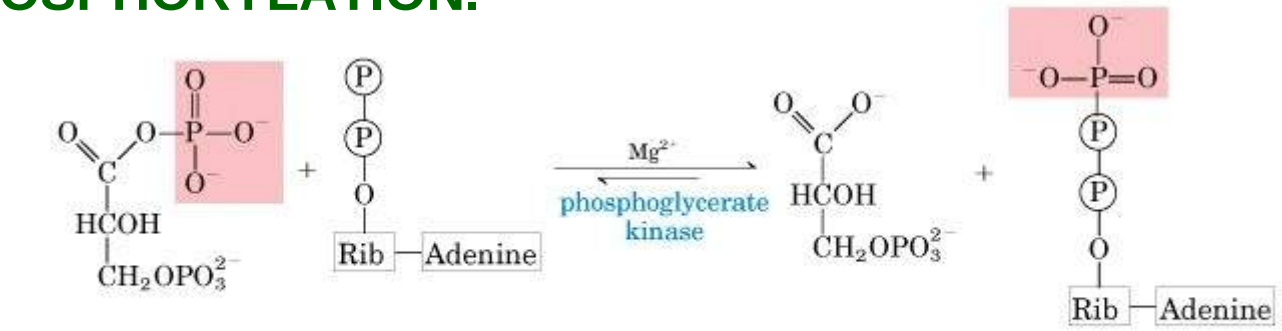
7. Phosphoglycerate kinase Reaction: Transfer of phosphoryl group fron 1,3 bisphosphoglycerate to ADP generating ATP.

I. The name of this enzyme indicates its function for reverse reaction.

II. It catalyses the formation by proximity effect. ADP-Mg bind on one domain and 1,3BPG binds on the other and a conformational change brings them together similar to hexokinase.

III. This reaction and the 6th step are coupled reaction generating ATP from the energy released by oxidation of 3-phosphoglyceraldehyde.

IV. This step generates ATP by SUBSTRATE-LEVEL PHOSPHORYLATION.



$\Delta G'^{\circ} = -18.5$ kJ/mol

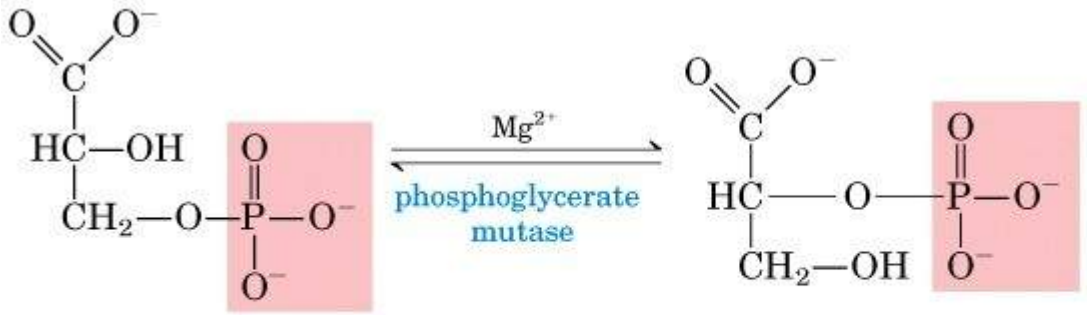
8. Phosphoglycerate Mutase Reaction: Conversion of 3-phosphoglycerate to 2-phosphoglycerate (2-PG).

I. In active form, the phosphoglycerate mutase is phosphorylated at His-179.

II. There is transfer of the phosphoryl group frm enzyme to 3-PG, generating enzyme bound 2,3-biphosphoglycerate intermediate. This compound has been observed occasionally in reaction mixture.

III. In the last step of reaction the phosphoryl group from the C-3 of the intermediate is transferred to the enzyme and 2-PG is released.

IV. In most cells 2,3BPG is present in trace amount, but in erythrocytes it is present in significant amount. There it regulates oxygen affinity to hemoglobin.



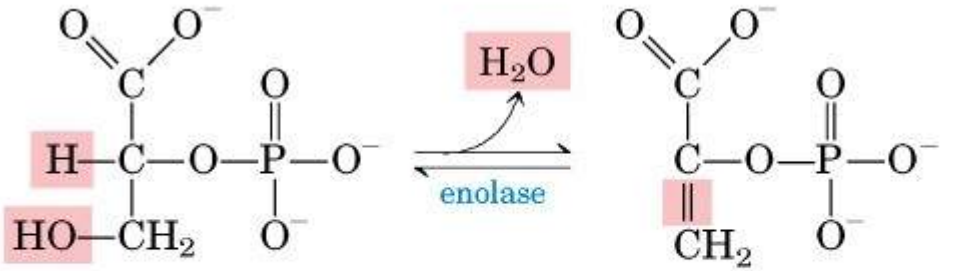
3-Phosphoglycerate

2-Phosphoglycerate

$\Delta G'^{\circ} = 4.4$ kJ/mol

9. Enolase Reaction: Dehydration of 2-phosphoglycerate (2-PG) to phosphoenolpyruvate (PEP).

I. Dehydration of 2-PG by this reaction increases the standard free enrgy change of hydrolysis of phosphoanhydride bond.
II. Mechanism: Rapid extraction of proton from C-2 position by a general base on enzyme, generating a carbanion. The abstracted proton is readily exchanges with solvent.
III. The second rate limiting step involves elimination of OH group generating PEP

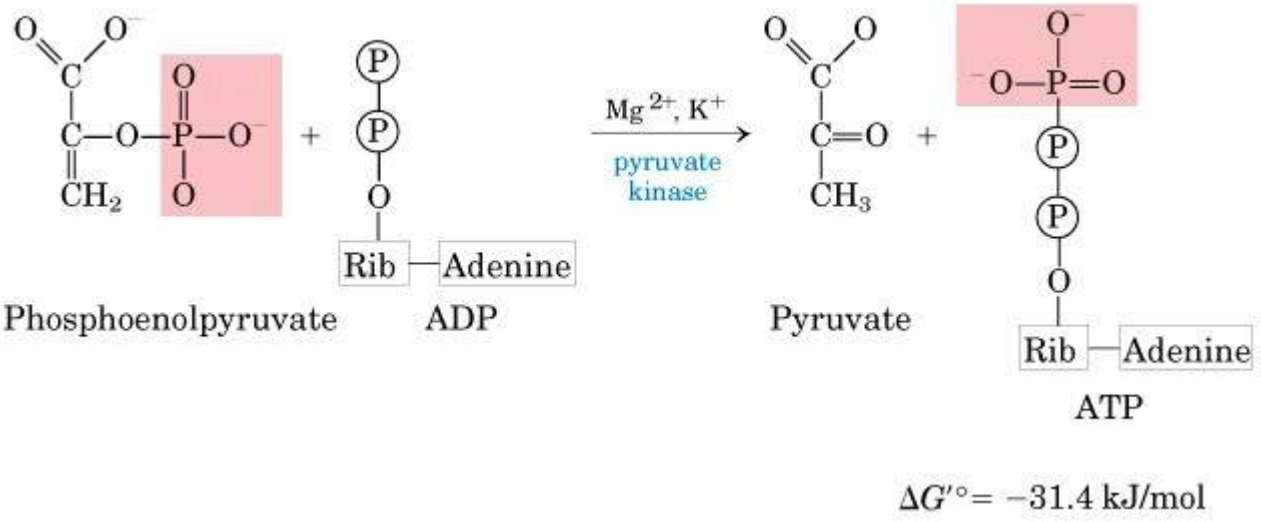


2-Phosphoglycerate Phosphoenolpyruvate

$\Delta G'^{\circ} = 7.5$ kJ/mol

10. Pyruvate Kinase Reaction: Transfer of phosphoryl group from PEP to ADP generating ATP and Pyruvate.

I. This is the second substrate level phosphorylation reaction of glycolysis.
II. This enzyme couple the free enrgy of PEP hydrolysis to the synthesis of ATP
III. This enzyme requires Mg^{++} and K^{+}



$\Delta G'^{\circ} = -31.4$ kJ/mol

Regulation of Glycolysis:

Two types controls for metabolic reactions:

a) Substrate limited : When concentrations of reactant and products in the cell are near equilibrium, then it is the availability of substrate which decides the rate of reaction.

b) Enzyme-limited: When concentration of substrate and products are far away from the equilibrium, then it is activity of enzyme that decides the rate of reaction. These reactions are the one which control the flux of the overall pathway.

There are three steps in glycolysis that have enzymes which regulate the flux of glycolysis.

I. **The hexokinase (HK)**
II. **The phoshofructokinase (PFK)**
III. **The pyruvate kinase**

Phosphofructokinase-1 (PFK-1):

It's activity is controlled by a complex allosteric regulation.
This reaction commits the cells to channel glucose to glycolysis.

ATP is the end product of glycolysis as well as it is substrate for PFK-1.
In presence of high concentration of ATP, ATP binds to inhibition site of PFK, and thereby decreases the activity of enzyme.

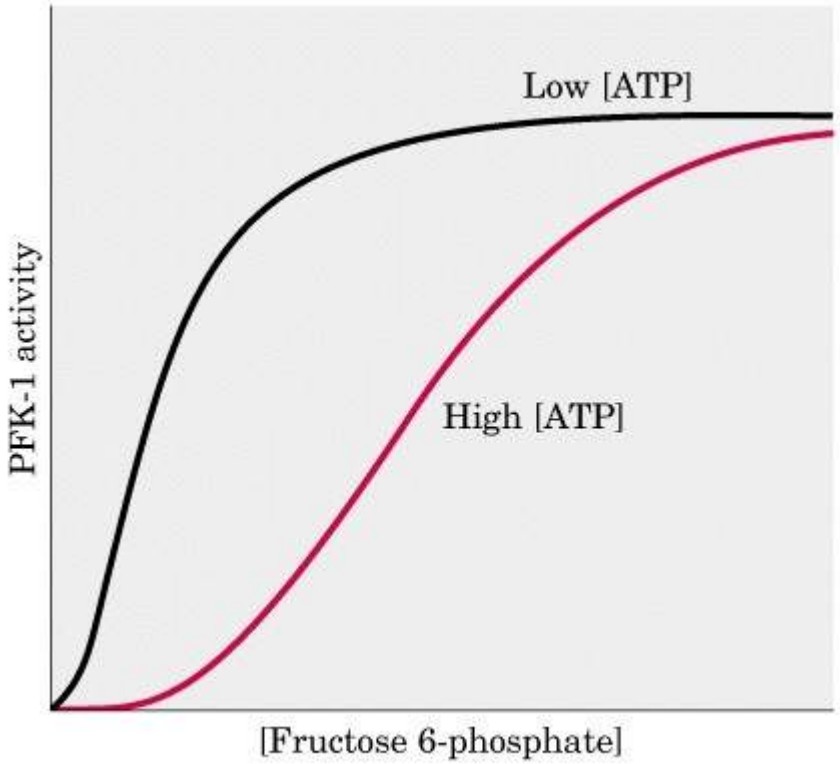
AMP, ADP and Fructose 2, 6 biphosphate act as allosteric activators of this enzyme.
Activation of enzyme by AMP overcomes the inhibitory effect of ATP.

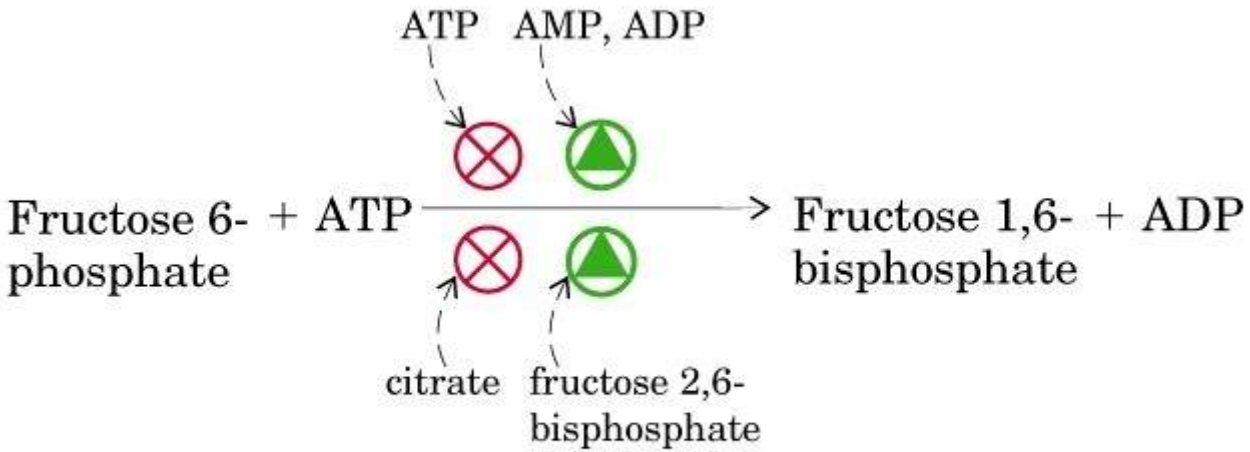
Two other enzymatic activities are involved in the regulation of PFK.

a) Adnylate kinase: It readily equilibrates 2 ADP molecules to one ATP and 1 AMP: $2ADP = ATP + AMP$, $K = [ATP][AMP] / [ADP] = 0.44$

Any decrease in ATP and increase in ADP results in an increase in AMP concentration, which activates PFK.

b) Fructose 1,6-bisphosphatase (FBPase): It catalyzes conversion of FBP to Fructose 6-phosphate, thus reverting back the PFK reaction.


Low [ATP]
High [ATP]
PFK-1 activity
[Fructose 6-phosphate]


ATP
AMP, ADP
Fructose 6-phosphate
+ ATP
Fructose 1,6-bisphosphate
+ ADP
citrate
fructose 2,6-bisphosphate

(c)

Substrate cycle or futile cycle: In order to control the flux of glycolysis and to have better regulation, cells have FBPase which keeps degrading the product of PFK reaction (FBP) to its substrate (F-6-P). This is called substrate cycle.

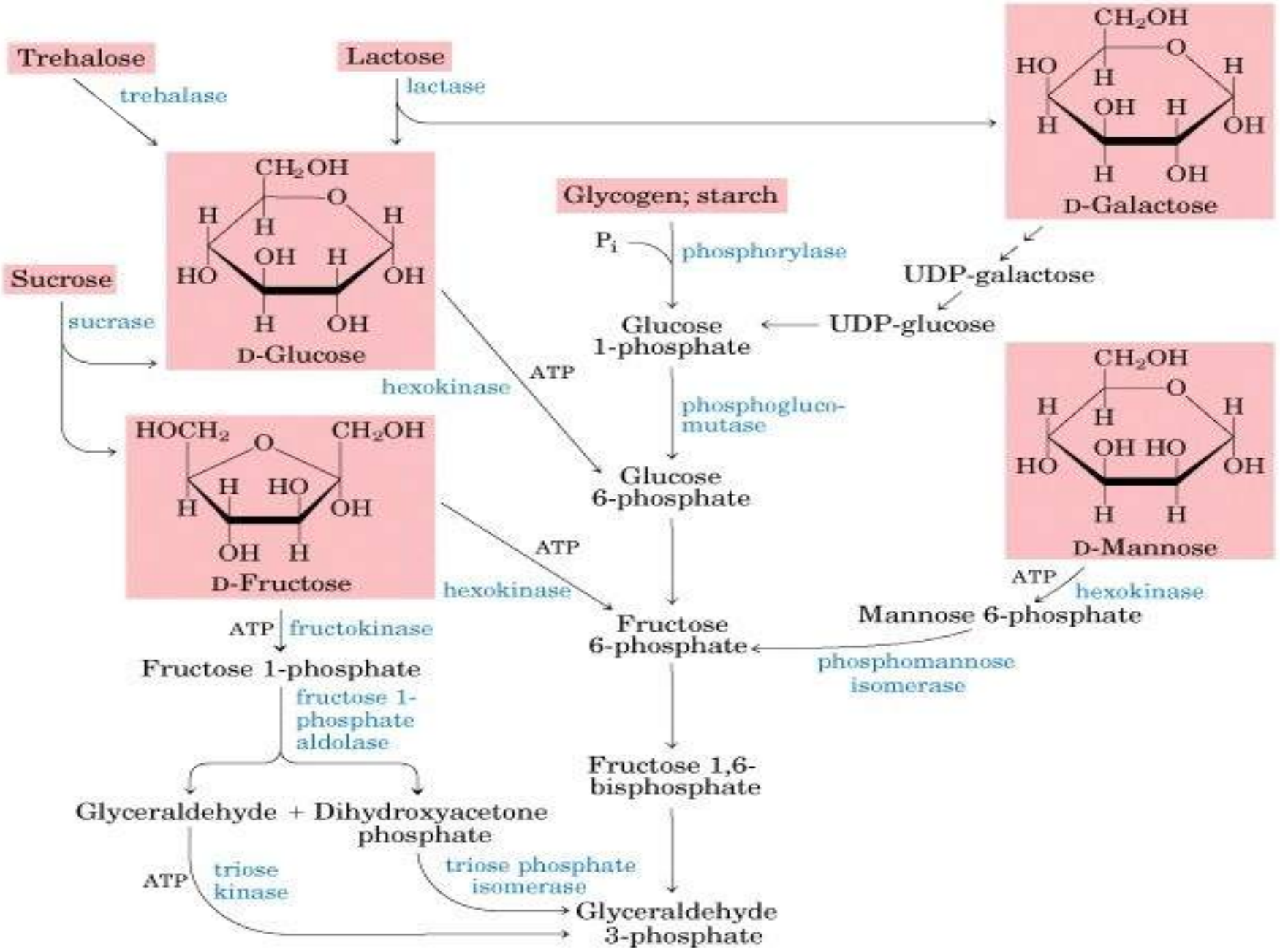
This is a futile exercise where, cells invest an ATP to produce FBP which is hydrolysed back to F6-P by FBPase. This is a price cells pay to keep glycolysis in check.

AMP acts as a potent inhibitor of FBPase. Thus the rate of glycolysis can be increased many fold by AMP as it activates PFK and at the same time it inhibits FBPase activity.

Hexokinase: It is allosterically inhibited by its product Glucose 6 phosphate. In liver Glucokinase is inhibited by Fructose 6 Phosphate. Uncomplexed ATP acts as a competitive inhibitor of this enzyme.

Pyruvate Kinase: It is allosterically inhibited by ATP. ATP binding to the inhibitor site of pyruvate kinase decxreases its ability to bing to phosphoenol pyruvate (PEP) the substrate.

It is also inhibited by Acetyl coenzyme A and long chain fatty acid.


Trehalose
trehalase
Lactose
lactase
D-Galactose
Glycogen; starch
P_i
phosphorylase
UDP-galactose
Sucrose
sucrase
D-Glucose
Glucose 1-phosphate
UDP-glucose
ATP
hexokinase
phosphogluco-mutase
Glucose 6-phosphate
D-Mannose
D-Fructose
ATP
hexokinase
ATP
hexokinase
Mannose 6-phosphate
ATP
fructokinase
Fructose 6-phosphate
phosphomannose isomerase
Fructose 1-phosphate
fructose 1-phosphate aldolase
Fructose 1,6-bisphosphate
Glyceraldehyde + Dihydroxyacetone phosphate
ATP
triose kinase
triose phosphate isomerase
Glyceraldehyde 3-phosphate